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Diving behaviour and prey of the Humboldt Penguin (*Spheniscus humboldti*)

Rory P. Wilson, Marie-Pierre Wilson, David Cameron Duffy, Braulio Araya M
and Norbert Klages

Underwater diving by volant seabirds is almost invariably concerned with foraging. Measurement of the dive duration can help elucidate underwater behaviour of seabirds; dive times positively correlated with depth indicate that the birds may be feeding on the bottom (e.g. DEWAR 1924) whereas dive times not correlated with depth, or very short in relation to depth, indicate that the birds are probably swimming near the surface or in mid-water regions (e.g. SIEGFRIED et al. 1975).

Penguins swim underwater to travel, to escape predators and to forage (e.g. KOOYMAN 1975). Although little is known about the diving behaviour of penguins, travelling birds appear to dive for relatively short periods, remaining near the surface (WILSON 1985a, TRIVELPIECE et al. 1986), whereas foraging birds generally dive for long periods (KOOYMAN et al. 1982, WILSON 1985a, TRIVELPIECE et al. 1986) and can dive deeply (e.g. KOOYMAN et al. 1982).

Whilst in Chile, in 1985, exceptional water clarity enabled us to observe the behaviour of Humboldt Penguins *Spheniscus humboldti* swimming underwater. We report how the underwater behaviour of these birds affected their dive times and we examine how this behaviour might enhance their chances of catching prey.

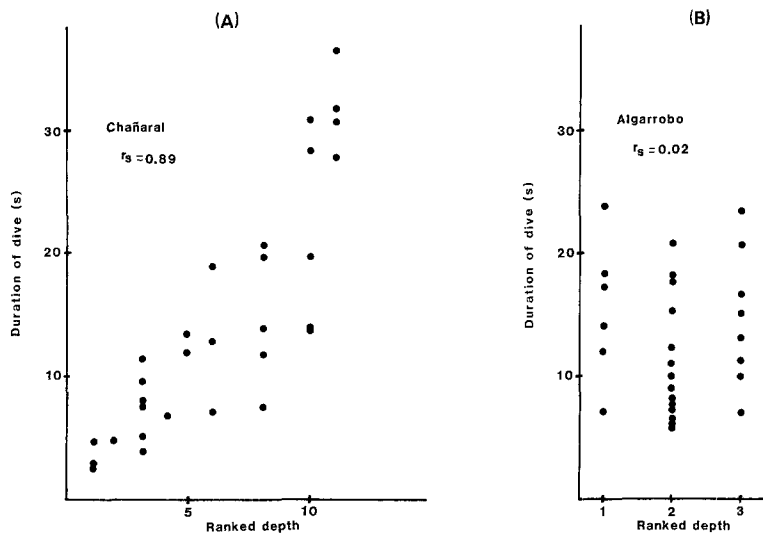
Methods

At Isla Chañaral (29°01'S, 71°50'W), between 31 October and 2 November 1985, we observed penguins leaving a small nesting colony (ca. 200 birds) and swimming out to see in groups of up to 10 individuals. They entered the water from a beach that sloped gently into the sea. We divided the water into 11 depth zones (maximum depth 30 m, based on coastal navigation charts) and noted the dive durations of all penguin groups in relation to the ranked depth of water at the midpoint of the dive. A similar procedure was used to observe Humboldt Penguins at Isla Algarrobo (33°30'S, 71°45'W), central Chile, between 6 and 8 November 1985. Here, near the departure beach there were three main water depths (shallow, <1 m; intermediate, 1–4 m; and deep, >4 m), which were marked by buoys.

Between 16 h 20 and 19 h 00, we caught penguins as they returned to their breeding colonies on the two islands. Sixteen birds caught at Chañaral and 18 at Algarrobo were stomach-pumped (WILSON 1984) and the samples sorted immediately. All prey were examined for marks made by the birds' beaks (WILSON & DUFFY 1986) and were ranked according to their state of digestion from 1 (freshly caught) through 6 (most digested, WILSON 1985b). Specimens were then preserved in alcohol and were identified later by comparison with reference specimens in the Port Elizabeth Museum, South Africa and by using ZAMA & CARDENA (1984) and CLARKE (1986).

Results and Discussion

The clarity of the water at Isla Chañaral enabled us to see the penguins during almost their entire dives (up to an estimated depth of 10–15 m). The penguins invariably dived to the seabed and then immediately returned to the surface. The dive and return-to-the-surface angles were both judged by eye to be about 45° . Dive times were longer for birds diving in deeper water ($r_s = 0.89$, $P < 0.001$, $n = 30$, Fig. a) and for those birds further from the shore (and presumably in deeper water). Dives in the two deepest zones (max. 30 m depth; $\bar{x} = 26.2$ s, $SD = 8.2$, $n = 14$) were significantly shorter than durations reported by DUFFY (1983) in deep water (>100 m) off Peru ($\bar{x} = 75.0$ s, $SD = 44.9$, $n = 28$; $t = 4.0$, $P < 0.01$).



Relationship between duration of dive and depth (ranked, not actual) for Humboldt Penguins at (a) Isla Chañaral and (b) Isla Algarrobo.

At Algarrobo, the water was more turbid and we were not able to see the penguins during all dives. Yet, most birds observed swam within a metre of the surface for the full duration of the dive. Here, there was no relationship between dive time and depth ($r_s = 0.02$, $P > 0.05$, $n = 39$, Fig. b). Mean dive time was 13.3 s ($SD = 5.2$).

Penguins at both Chañaral and Algarrobo fed predominantly on pelagic schooling fish. Prey items at Chañaral were garfish *Scomberesox spp* (94 %, by number), anchovy *Engraulis ringens* (3 %) and sardine *Sardinops sagax* (1 %). At Algarrobo, penguins fed on anchovy (72 %), sardine (4 %) and squid *Todarodes fillippovae* (14 %). Fish caught at Chañaral were significantly less digested than fish caught at Algarrobo (mean states of digestion of 3.95, $SD = 1.58$, $n = 43$ and 5.05, $SD = 0.64$, $n = 64$, respectively; $t = 5.00$, $P < 0.01$). Fifteen of sixteen of the freshly caught fish from both localities had been seized by the penguin from below since the tomial incisions ran dorsoventrally and were closest to each other on the dorsal aspect of the fishes (cf. WILSON & DUFFY 1986).

Although penguins in both localities left their islands after sunrise, Chañaral birds appeared to forage much closer to the colony; they swam slowly out to sea, were visible near the island throughout the foraging period and returned with relatively undigested prey. Algarrobo penguins swam comparatively rapidly away from the island and were usually lost from view within ten minutes of leaving the beach. For much of the day, no penguins could be seen from the island and returning birds contained well digested prey.

The short, shallow dives of the Algarrobo birds were typical of travelling penguins (cf. WILSON 1985a, TRIVELPIECE et al. 1986). Such shallow dives would be the most efficient form of horizontal travel because no time is lost in vertical movement. Penguins normally only swim at one (presumably optimal) speed (WILSON 1985a, ADAMS 1987, ADAMS & WILSON 1987), so distance covered is directly proportional to the time spent underwater. Birds which bounce dive cannot travel horizontally effectively because the distance covered is proportional to the dive depth multiplied by twice the tan of the dive angle. TRIVELPIECE et al. (1986) report that foraging Chinstrap *Pygoscelis antarctica* and Gentoo Penguins *Pygoscelis papua* dive for longer periods than travelling birds, but do not swim as far horizontally. Thus, it is likely that the bounce-diving Humboldt Penguins were also foraging. Future studies examining the horizontal distance travelled in relation to dive time (eg. WILSON 1985a) would clarify whether bounce diving occurs in other species of penguin.

The Humboldt Penguin is similar to its conspecifics in feeding primarily on pelagic school fish (BOERSMA 1976, WILSON 1985c, SCOLARO & BADANO 1986) which inhabit the upper water layers (eg. JOHANESSON & VILCHEZ 1980). At least two of the four species of *Spheniscus* penguin are known to capture prey from below (WILSON & DUFFY 1986). Pelagic school fish have a complex arrangement of light-reflecting guanine and hypoxanthine crystals embedded in their scales which ensures that they are minimally visible when viewed from almost any angle (DENTON & NICOL 1965). However, this system is not effective when the fish is perceived from directly below because the dim light reflected by the fish's ventral surface is far less intense than the background light impinging from the sea surface (DENTON 1971). Humboldt Penguins have an optical system designed to perceive objects above them best because the maximum width of their retinal binocular field is above the line of the bill and the vertical extent of the binocular field extends 125° (MARTIN & YOUNG 1984). We suggest that the bounce diving behaviour of the Humboldt Penguin enhances the chances of prey capture because the birds spends much of its time deep in the water column where it is most likely to perceive surface-swimming fish.

This paper is based on results of the FitzPatrick Institute's 25 anniversary Expedition to Chile. We are grateful to R. KELLY and the Confradia Nautica de Pacifico for access to Algarrobo and to F. BUCHHOLZ and B. CULIK for help with the manuscript.

Summary

Humboldt Penguins diving in clear water in Chile had two distinct behaviours; short dives ($\bar{x} = 13.3$ s) where the birds remained within a metre of the surface and bounce dives, where the penguins descended the water column to the sea bed where they immediately returned to the surface. Here, dive time was correlated with water depth. Stomach pumped penguins had eaten predominantly pelagic school fish, *Scomberesox*, *Engraulis* and *Sardinops*. Most prey (94 %) had been seized from below. We suggest that short, shallow dives are typical of travelling penguins. Bounce dives, however, would enable foraging penguins, which have an enhanced binocular field above the line of the bill, to perceive surface-swimming prey more easily because silvery clupeid fish appear as silhouettes.

Zusammenfassung

In klaren Gewässern in Chile tauchende Humboldt-Pinguine zeigten zwei deutlich verschiedene Verhaltensweisen. Während kurzer Tauchgänge ($\bar{x} = 13,3$ s) blieben die Tiere im Bereich von 1 m unter der Oberfläche. Tauchgänge durch unmittelbares Abtauchen in einem Winkel von 45° eingeleitet, führten direkt zum Meeresboden; die Vögel kehrten sofort wieder an die Oberfläche zurück. In diesem Fall war die Tauchzeit von der Wassertiefe abhängig. Pinguine, deren Magen durch Auspumpen untersucht wurde, hatten vorwiegend pelagische Schwarmfische gefressen: *Scomberesox*, *Engraulis* und *Sardinops*. Die Mehrzahl der Fische (94 %) wurde von unten kommend erfaßt. Nach unserer Auffassung sind die kurzen und flachen Tauchgänge typisch für nicht jagende Pinguine, die längere horizontale Strecken zurücklegen. Das tiefe Abtauchen dagegen ermöglicht jagenden Pinguinen, die ein besonders gutes Fernsichtvermögen oberhalb des Schnabels besitzen, nahe der Oberfläche schwimmende Beute leichter zu erfassen. Die sonst silbrigen clupeiden Fische sind als dunkle Silhouetten gegen die Wasseroberfläche besser zu erkennen.

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Author's addresses: (RPW, M-PW, DCD) Percy FitzPatrick Institute of African Ornithology, University of Cape Town, Rondebosch, 7700, South Africa; (BAM) Instituto de Oceanología, Universidad de Valparaiso, Casilla 13-D, Viña del Mar, Chile; (NK) Port Elizabeth Museum, Port Elizabeth, South Africa.

Present address: (RPW) Institut für Meereskunde an der Universität Kiel, Düsternbrooker Weg 20, D-2300 Kiel 1, F.R.G.

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Rückstände chlororganischer Verbindungen in Gelegen des Triels (*Burhinus oedicnemus*) im Elsaß

Markus Nipkow

Einleitung

Seit Mitte der siebziger Jahre bestehen in den meisten europäischen Ländern Anwendungsverbote für das Insektizid DDT, den bekanntesten Vertreter der umwelttoxikologisch besonders bedenklichen chlorierten Kohlenwasserstoffe, sowie einige andere Pestizide aus dieser Gruppe. In der Zwischenzeit konnte an verschiedenen Vogelarten eine deutliche Abnahme der Kontamination mit bestimmten chlororganischen Verbindungen festgestellt werden (z. B. BAUM & CONRAD 1978; BECKER, CONRAD & SPERVELAGE 1980; BECKER, BÜTHE & HEIDMANN 1985, 1988; HEIDMANN et al. 1988). Andererseits zeigen gerade auch jüngste Untersuchungen keinen deutlichen Trend beim DDE (Hauptmetabolit des DDT) und weisen wieder vermehrt Rückstände an unmetabolisiertem DDT nach (BURGERS et al. 1986; OLSSON & REUTERGARDH 1986). Besonders die weit ziehenden Arten laufen Gefahr, im Winterquartier und auf den Zugwegen mit Schadstoffen belastet zu werden, da der weltweite Pestizidverbrauch kontinuierlich im Steigen begriffen ist: allein von 1965 bis 1980 von 2,75 Mio. Tonnen auf 4,57 Mio. Tonnen (PREDICASTS, in: Die Grünen im Bundestag/AK Umwelt 1985).

Bei einer Habitatwahl-Untersuchung an der elsässischen Trielpopulation (NIPKOW 1987), die ausschließlich landwirtschaftliches Kulturland besiedelt, ergab sich die Gelegenheit zu einer Analyse von Eiern auf Pestizidrückstände.

Material und Methode

1986 wurden 10 Eier aus 6 Gelegen der Trielpopulation im Elsaß (Département Haut-Rhin, Frankreich) am Tierhygienischen Institut in Freiburg auf Pestizidrückstände untersucht. 7 Eier stammten von spät gezeitigten Ersatzgelegen; solche werden grundsätzlich nicht weiter bebrütet, wenn die das Nest umgebende Vegetation eine bestimmte Höhe überschreitet (NIPKOW 1988). Zwei weitere Eier wurden nach maschineller Feldbearbeitung verlassen aufgefunden. Ein Ei stammte aus einem 2er-Gelege, bei dem das andere erfolgreich ausgebrütet worden war.